

Simulation Discussion + Short Lab Update

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Master Thesis Progress Report

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QuTip Hamiltonians

$$H = H_0 + H_{JC} + H_{drive} \cdot f(t)$$

Bare Energies Drive Operator

Jaynes-Cummings Gaussian Pulse

- $H_0 = \Delta\omega_c a_\pi^\dagger a_\pi + (\Delta\omega_c + 0.5) a_v^\dagger a_v + \Delta_a \sigma^\dagger \sigma$
- $H_{JC} = g \cdot (a_\pi \sigma^\dagger + a_\pi^\dagger \sigma + a_v \sigma^\dagger + a_v^\dagger \sigma)$
- $H_{drive} = i\mu_{fc} \sqrt{\kappa_{oc}} (a - a^\dagger)$
- $f(t) = A \cdot e^{-\frac{(t-t_0)^2}{2\sigma^2}} = a_{in}(t)$

Input-Output Relation

$$a_{\text{out},\{\pi,\nu\}} = \mu_{\text{fr}} a_{\text{in}}(t) + \mu_{\text{fc}} \sqrt{\kappa_{\text{oc}}} \langle a_{\{\pi,\nu\}} \rangle(t)$$

1. Mode Overlap

$$c_{\{\pi,\nu\}} = \int a_{\text{ideal}}^*(t) \cdot a_{\text{out},\{\pi,\nu\}}(t) dt$$

2. State Analysis

$$A_{\pm} = \frac{c_{\nu} \pm c_{\pi}}{\sqrt{2}}$$

$$P_{\pm} = |A_{\pm}|^2$$

Fidelity Calculation Methods

1. Pure Simulation (F_{sim})

$$F_{\text{pure}} = F_{\text{sim}}$$

2. Analytical Model

Get F_1 from sim at very low $\bar{n}$.
Account for multi-photon events:

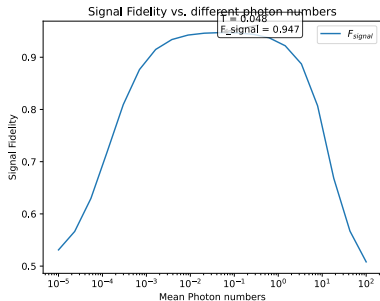
$$F_{\text{ana}} = 0.5 + (F_1 - 0.5)e^{-(1-\eta)\bar{n}}$$

3. Hybrid: Sim + Analytical

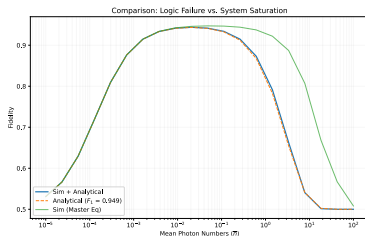
Extract F_{sim} from simulation, then
apply analytical scaling to account for
multi-photon contributions:

$$F_{\text{hyb}} = 0.5 + (F_{\text{sim}} - 0.5)e^{-(1-\eta)\bar{n}}$$

Results: Fidelity and Atomic Scattering vs Photon Number



Fidelity vs $\langle n \rangle$



Comparison between different ways of
computing Fidelity

Gate Characterization: CPHASE (H/V Basis)

Process Matrix χ in the H/V Basis

$\langle \text{out} \text{in} \rangle$	$ 0\pi\rangle$	$ 0V\rangle$	$ 1\pi\rangle$	$ 1V\rangle$
$ 0\pi\rangle$	0.9440	0.0177	0.0000	0.0000
$ 0V\rangle$	0.0560	0.9823	0.0000	0.0000
$ 1\pi\rangle$	0.0000	0.0000	0.8878	0.0178
$ 1V\rangle$	0.0000	0.0000	0.1122	0.9822

Gate Fidelity

$$F = 94.91\%$$

Gate Characterization: CNOT (R/L Basis)

Process Matrix χ in the Circular Basis

$\langle \text{out} \text{in} \rangle$	$ 1R\rangle$	$ 1L\rangle$	$ 0R\rangle$	$ 0L\rangle$
$ 1R\rangle$	0.7387	0.2593	0.0000	0.0000
$ 1L\rangle$	0.2613	0.7407	0.0000	0.0000
$ 0R\rangle$	0.0000	0.0000	0.2598	0.8102
$ 0L\rangle$	0.0000	0.0000	0.7402	0.1898

Gate Fidelity

$$F = 75.75\%$$

Gate Characterization: CNOT (A/D Basis)

Process Matrix χ in the Diagonal Basis

$\langle \text{out} \text{in} \rangle$	$ 1A\rangle$	$ 1D\rangle$	$ 0A\rangle$	$ 0D\rangle$
$ 1A\rangle$	0.8476	0.4966	0.0000	0.0000
$ 1D\rangle$	0.1524	0.5034	0.0000	0.0000
$ 0A\rangle$	0.0000	0.0000	0.2370	0.9009
$ 0D\rangle$	0.0000	0.0000	0.7630	0.0991

Gate Fidelity

$$F = 75.37\%$$